



Pittala Sree Sai Pavan

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B.Tech – Electronics and Communication Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/%	Year
B.Tech.	Indian Institute of Technology Guwahati	7.13 (Current)	2024 – Present
Senior Secondary	Telangana Board of Intermediate Education	96.2%	2024
Secondary	Telangana Board of Secondary Education	9.7	2022

PROJECTS

- CVM++ — Stack-Based Virtual Machine & Compiler** Jan. 2026 – Apr. 2026
IIT Guwahati Coding Club — Even Semester Projects 2026 (Team) GitHub
 - Built a C++17 Lexer → Parser → Bytecode Compiler → Stack VM pipeline with zero external dependencies.
 - Designed a recursive-descent parser with 6-level precedence, operator associativity and a 12-node typed AST.
 - Engineered a Lexer tokenizing 25+ token types with line-tracked error and graceful recovery across all stages.
 - Supported variables, arithmetic, booleans, control flow, and I/O via a fetch-decode-execute stack execution model.
 - Implemented a 23-opcode bytecode ISA with 2-pass forward-jump patching, label resolution, and disassembler.
- MWIS Solver — Maximum Weight Independent Set** Jan. 2025 – Apr. 2025
IIT Guwahati Coding Club — Even Semester Projects 2025 (Team) GitHub
 - Built a 3-phase solver: branch-and-bound DFS with suffix-sum pruning over top-24 skill-to-degree ranked nodes.
 - Used 32-bit adjacency bitmasks for $O(1)$ conflict detection on graphs up to 200,000 nodes and $O(N^2)$ edges.
 - Applied `getchar_unlocked` buffered I/O achieving 10–100× faster input parsing than `cin`.
 - Boosted weight 10–50% over greedy via ADD/SWAP local search with $\pm 5\%$ perturbed restarts within 300s.
- GitLite — Git-Inspired Version Control System** 2025 – Present
Personal Project (Ongoing) GitHub
 - Engineered a fully functional Git-inspired VCS from scratch in modern C++17 with no external dependencies.
 - Used OpenSSL SHA-256 hashing to build a content-addressable object store for blobs, trees, and commit objects.
 - Implemented a commit DAG with symbolic HEAD refs and named branch pointers for full history traversal.
 - Built LCS-based line diffing using bottom-up dynamic programming and recursive backtracking between files.
 - Designed a plain-text staging index that hashes each file's content and clears atomically on every successful commit.

TECHNICAL SKILLS

- Programming:** C, C++, Python*
- Web Technologies:** HTML, CSS, JavaScript, React*
- Tools/Frameworks:** NumPy, Pandas, Matplotlib, SymPy, NetworkX*, Git*, GitHub*
- Operating Systems:** Windows*, Linux * Elementary proficiency

ACHIEVEMENTS

- Competitive Programming**, Max Rating **1404** on Codeforces (**Specialist**); Handle - **ssp_09** 2025 – 26
 - Ranked **1320** globally among 34,550+ participants in Codeforces Round 1090 (Div. 4).
 - Achieved Global Rank **1994** among 21,113+ participants in Codeforces Round 1083 (Div. 2).
- JEE Advanced 2024**, Ranked in the top **2.53%** among **1.8 lakh+** shortlisted candidates across India 2024
- TS EAPCET 2024**, Ranked in the top **0.87%** among **2.4 lakh+** candidates across Telangana State 2024

KEY COURSES TAKEN

- Computer Science:** Introduction to Computing, Computing Laboratory
- Mathematics:** Linear Algebra, Calculus, Differential Equations, Real Analysis, Complex Analysis
- Core Engineering:** Basic Electronics, Digital Circuits (Theory & Lab), Circuit Theory, Signals and Systems

EXTRACURRICULAR ACTIVITIES

- PR & Branding Associate**, Alcheringa — Northeast India's largest cultural fest Nov. 2024 – Feb. 2025
- Junior Web Developer**, Students' Web Committee — frontend dev on institute projects Oct. 2025 – Present
- HabitFlow**, Personal project — React habit tracker with streaks, analytics & progress visualization Oct. 2025
- NSS Volunteer**, NSS Cell, IIT Guwahati — 30+ hours of community service and outreach activities Jan. 2026 – Present